

Importance of conductivity, periodicity and collimation in a model for transmission of energy through sub-wavelength, periodically arranged holes in a metal film

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We investigate the difference between theoretical predictions and experiments measuring the transmission of energy through sub-wavelength, periodically arranged holes in a metal film. At normal incidence, our theoretical model predicts zero transmission when the wavelength is equal to the periodicity, and 100% transmission at two nearby wavelengths. Experiments, however, observe a less sharp minimum and only one transmission maximum. By incorporating each feature into our model, we find that varying conductivity and abandoning strict periodicity does not account for these differences, while imperfect conductivity contributes significantly.

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1. Introduction

The subject of extraordinary optical transmission from sub-wavelength-size holes is well documented, beginning with the work of Ebbessen et al. in 1998 [1]. The transmission of energy experiences a minimum when the wavelength of the incident energy is equal to the period width (as well as other wavelengths), and one or more maxima at nearby wavelengths. Numerical and physical experiments have explored many different hole shapes, hole arrangements, and metallic compositions. However, theoretical and experimental results consistently differ from each other in the number and shape of transmission maxima. In the case of a two-dimensionally periodic square array of holes in a perfect conductor, [2–4] show multiple narrow transmission maxima of unit height at wavelengths close to the period width, while experimental results show one broader peak [1, 5, 6].

In this paper we consider a typical model for computing transmission and three of its features that cause it to differ from experiments. The first is the finite conductivity of a real metal, which we show contributes very little to the degradation of the twin maxima. This is supported by simulations in COMSOL Multiphysics and Microwave Studio, and we find that our model is much more efficient than these more traditional methods. The second is the difference between an infinitely periodic array of holes and a finite array, which we show causes a change in the total transmission but not a qualitative change in the transmission. Finally we allow for imperfect collimation of the incident radiation. We find that this causes both a decrease in transmission and a qualitative change in the transmission profile, in a way that is consistent with the experimental results.

We begin by representing the electromagnetic (EM) fields outside the metal film as a series of plane waves, and inside the holes by a series of waveguide modes. We then determine the amplitude coefficients of each mode by applying the matching conditions associated with Maxwell’s equations at each side of the slab. This method has several advantages over more traditional numerical techniques. First, in many cases only a few waveguide modes contribute significantly to the EM field, which greatly reduces calculation time. Second, separating the field into individual modes highlights the emergence of the Wood anomaly, a phenomenon that will be described below. Finally, corners and resonances cause no extra difficulty for the mode-matching technique, while we find that COMSOL and Microwave Studio require extraordinary amounts of time and memory to resolve the fields in these circumstances.

We begin Section 2 with a periodic array of holes in a perfect conductor, illuminated on one side by a plane wave. In Section 3 we introduce finite conductivity into our equations; this involves exact evaluation of the waveguide modes within the holes and extending into the metal with an asymptotic calculation of the propagation constant. This is supported by COMSOL. In Section 4 we approximate a finite array of holes by forming supercells containing multiple holes surrounded by a large amount of space with no holes. Finally, in

Section 5 we allow the collimation of the incoming radiation to include a range of angles, using an averaging technique that involves no extra approximation.

2. A Perfectly Conducting, Periodic Film Illuminated by a Plane Wave

In this section we consider an array of holes that is periodic in the x - and y - directions in a perfectly conducting metal film with thickness h in the z -direction, although the same formalism may be used to evaluate the case of a one-dimensionally periodic array of slits, as we do in Sec. 4. (See Fig. 1.) We draw from the papers of Martin-Moreno and Garcia-Vidal et al. [2, 3], although similar work has been done by other authors (for example, [4]).

The role of periodicity is two-fold. First, it allows us to invoke Bloch theory, which tells us that solutions of Maxwell's equations in a periodic system will be Bloch periodic, implying that we need only consider one cell period in our calculations. This also implies that solutions outside the slab can be written as a *countable* combination of plane waves, rather than the continuous sum that is required in the nonperiodic case. Meanwhile, inside the holes, a countable basis for solutions is given by the waveguide modes, so the matching process reduces to an infinite matrix equation rather than an integral equation. Second, periodicity causes the Wood anomaly, which occurs when one of the plane waves outside the film becomes parallel to the surface of the film. This causes the transmission of energy into and through the holes to be zero (and thus total reflection from the film).

The Wood anomaly is straightforward to predict. If the film is periodic in the x - and y - directions and z is the direction of propagation, an EM field in free space is composed of countably many plane waves, which are the product of a constant vector times the exponential

$$\exp i(k_x x + k_y y + k_z z), \quad (1)$$

where $k_z^2 = \omega^2 - k_x^2 - k_y^2$ to satisfy Maxwell's equations for frequency ω . The wavevector (k_x, k_y, k_z) encodes the angle of incidence, and the values it may take are determined by the periodicity and the frequency. (The plane wave is so called because its level-sets are planes of the form $k_x x + k_y y + k_z z = b$, where b is a constant. Thus $k_x = k_y = 0$ corresponds to normal incidence.) The Wood anomaly occurs when $k_z = 0$.

The reason why the Wood anomaly causes total reflection is also plain. Plane waves come in two varieties: p -polarized, for which $H_z = 0$, and s -polarized, for which $E_z = 0$. The constant coefficients of the $\mathbf{E}$ and $\mathbf{H}$ components of the plane wave are related by

$$\begin{pmatrix} H_y \\ -H_x \end{pmatrix} = \pm Y \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \quad (2)$$

where the *admittance* Y is given by $Y = \omega/k_z$ for p -polarized waves and $Y = k_z/\omega$ for s -

polarized waves. In the condition of a Wood anomaly $k_z = 0$ the admittance of the p -polarized wave becomes infinite, which produces zero transmission during the matching procedure.

Waveguide modes inside the holes also satisfy Eq. (2). When bounded by a perfect conductor, the waveguide modes also come in two varieties: transverse electric (TE) modes for which $E_z = 0$ (that is, the electric field is transverse to the direction of propagation) and TM modes for which $H_z = 0$. The propagation constant k_z of a waveguide mode depends on the geometry of the hole, the admittance of a TE mode is $Y = k_z/\omega$, and the admittance of a TM mode is $y = \omega/k_z$.

For complete details of the mode-matching procedure, consult Appendix I or [2, 3]; here we describe it only in broad terms. The EM fields are written as a series of plane waves to the left and right of the film and as a series of waveguide modes inside the holes, with coefficients representing the amplitude of each mode. The matching conditions associated with Maxwell's equations for a perfect conductor require that the components of E transverse to an interface be continuous and, except for air-metal interfaces, those of M as well. Thus, we set the transverse components of the three series equal at the interfaces and, taking advantage of the orthogonality of modes, apply inner products to single out each mode. Finally, the coefficients are written as the solution of an infinite matrix equation, which may be truncated and solved.

An example transmission profile produced by this procedure is shown in Fig. 2a for the case of normal incidence. The Wood anomaly seen at $\lambda/L = 1$ has associated with it two transmission maxima of unit height; the maximum closer to the Wood anomaly is thinner than the maximum further away. The other Wood anomaly pictured here at $\lambda/L = (\frac{1}{2})^{1/2}$ also has associated transmission maxima, though they are not so pronounced. The heights and precise locations of these maxima are not so easily predicted as the Wood anomaly and must be found, at least in part, numerically. Figure 2b shows the case of a p -polarized plane wave incident at a 5 degree angle, which demonstrates the birth of new Wood anomalies which join and separate as the angle changes.

We now proceed to alter our model to account for the finite conductivity of a real metal.

3. Finite Conductivity

Many commonly used metals have high conductivity and behave similar to a perfect conductor. However, the calculation of EM fields surrounding a finite conductor is significantly more complicated and requires some approximation. One method is discussed, for example, in [7, 8] Here, we propose a new method for the case of large but finite conductivity that is exact except for one asymptotic calculation.

Our mode-matching method requires explicit calculation of the waveguide modes inside the holes, which limits the hole shapes that we are able to consider. For the case of finite

conductivity we use circular holes. The explicit form of a single waveguide mode can be found, for example, in [9, 10]. However, two details remain.

The first detail is the overlap of waveguide modes between neighboring holes. In a good conductor, however, this feature is negligible, because the field decays quickly and exponentially inside the metal. The second, more pressing detail is the calculation of the propagation constant k_z of the waveguide modes. For real dielectric constant ϵ_1 inside the holes and real dielectric constant ϵ_2 in the metal, it is known that the propagation constant of each waveguide mode can be bounded and ordered [9, 10], but for complex ϵ_2 the propagation constant is also complex. The propagation constant is the solution to the equation [9]

$$\left(\frac{J'_n(a\lambda_1)}{a\lambda_1 J_n(a\lambda_1)} + \frac{K'_n(a\lambda_2)}{a\lambda_2 K_n(a\lambda_2)} \right) \left(\omega^2 \mu \epsilon_1 \frac{J'_n(a\lambda_1)}{a\lambda_1 J_n(a\lambda_1)} + \omega^2 \mu \epsilon_2 \frac{K'_n(a\lambda_2)}{a\lambda_2 K_n(a\lambda_2)} \right) = n^2 k_z^2 \left(\frac{1}{(a\lambda_1)^2} + \frac{1}{(a\lambda_2)^2} \right)^2, \quad (3)$$

where J_n and K_n are the Bessel functions, a is the radius of the circular hole, $\lambda_1 = \omega^2 \mu \epsilon_1 - k_z^2$, $\lambda_2 = k_z^2 - \omega^2 \mu \epsilon_2$, and the magnetic constant is assumed to be μ everywhere.

Equation (3) cannot be solved for k_z analytically and it poses numerical challenges as well. It may, however, be solved asymptotically. A perfect conductor is characterized by the dielectric constant $|\epsilon| = \infty$. More precisely, real metals have dielectric constant approaching $\epsilon = i\infty$ in the frequency range in question. By taking $1/\epsilon_2$ approaching zero and the propagation constant k_z approaching that of the associated waveguide mode in a perfect conductor k_0 , we find that

$$k_z \sim k_0 + \frac{C_1}{\epsilon_2^{1/2}} + \frac{C_2}{\epsilon_2} \quad (4)$$

to order $1/\epsilon_2$, where C_1, C_2 are complicated constants described in Appendix II.

Once we have explicit expressions for the modes, the rest of the mode-matching procedure entails only slight modification. Waveguide modes in a finite conductor no longer separate into TE and TM modes, instead becoming hybrid modes that have elements of both. Each mode has four different admittances: the TE and TM parts of the mode inside the hole, and the TE and TM parts inside the metal. The analogue of Eq. (2) for the case of finite conductivity is

$$\pm \begin{pmatrix} H_y \\ -H_x \end{pmatrix} = Y_{hole}^{TE} \begin{pmatrix} E_x \\ E_y \end{pmatrix}_{hole}^{TE} + Y_{hole}^{TM} \begin{pmatrix} E_x \\ E_y \end{pmatrix}_{hole}^{TM} + Y_{metal}^{TE} \begin{pmatrix} E_x \\ E_y \end{pmatrix}_{metal}^{TE} + Y_{metal}^{TM} \begin{pmatrix} E_x \\ E_y \end{pmatrix}_{metal}^{TM}. \quad (5)$$

Finally, the matching conditions for Maxwell's equations are also different in the case of finite conductivity. In this case the components of both E and H tangential to all interfaces

must be continuous. This allows the field to penetrate into the top and bottom of the film as well as into the sides of the holes. The change in the calculations, however, is minimal, and is detailed in Appendix I.

A sample transmission profile using the same parameters as Fig. 2 is shown in Fig. 3 for two different values of ϵ_2 . The Wood anomaly is present in the same location as it was previously, and the location of the transmission maxima have changed little. For the smaller of the two dielectric constants considered, the transmission is slightly lower. For the larger of the two dielectric constants, the transmission profile is indistinguishable from that of the perfect conductor.

In parallel, we performed the same simulation numerically in COMSOL Multiphysics, a finite-element-based commercial EM software. The geometry of the model used in COMSOL is shown in Fig. 4a: a plane wave is incident from one boundary plane and the transmitted energy is collected on the other. The boundary conditions are periodic, since incident energy is normal to the film. Two different films are simulated: a perfect conductor (PEC), and Al at frequency 10 GHz.

As seen in Fig. 4b, the numerical simulation does not reproduce the unit height of the narrow transmission maximum, although it does reproduce the broad maximum. More importantly, however, the perfectly conducting metal and the Al film exhibit indistinguishable transmission profiles.

The disagreement between the numerics and our theory is due to the finite mesh and imperfect absorption at the radiation boundary plane in the simulator. A denser mesh helps the numerical results to match better with the theoretical prediction at the cost of more computing resources. The mesh used to create Fig. 4 has more than 170,000 mesh elements and creates more than 1 million degrees of freedom, which requires more than 32GBytes of memory. Secondly, the imperfect absorption boundary properties at the incident and evaluation planes generate extra oscillations not pictured here.

Similarly, we perform the simulation in Microwave Studio, in which we have better control of the reflection at the boundary. The additional ripples in the transmission profile can be eliminated when the reflection at the boundary is lower than -20dB. However, we still do not find perfect agreement at the Wood anomaly and the resonance.

4. Approximation of a Finite Array

Our mode-matching method inherently relies on periodicity, but we may simulate the behavior of a finite conductor in the following way. We define a large supercell consisting of a block of holes followed by a span of metal with no holes and repeat this periodically. As the amount of space between blocks tends to infinity, each block of holes will approach the behavior of a finite array. This method is computationally expensive but requires very little

change to the theoretical process.

In Fig. 5, we demonstrate this in the case of a one-dimensionally periodic array of infinite slits, which has a simpler transmission profile. In the one-dimensional case with no supercells (left), there is only a single transmission peak near the Wood anomaly at $\lambda = L$ rather than the double peak of the square array. We introduce finite arrays of 10 holes a distance of 10 array widths apart (center) and 50 array widths apart (right). We find that the qualitative behavior of the transmission profile does not change. The amount of transmission decreases by about 20%. There are some Fabry-Perot-like oscillations near the primary transmission maximum, which are due to the interaction between neighboring arrays and are thus an artifact of our approximation method. The same behavior is seen in arrays of as few as four holes. Therefore we find that a finite array of holes exhibits the same behavior as an infinite array, except that the total transmission is somewhat reduced.

5. Imperfect Collimation

In our basic method, the metal film is illuminated on one side by a plane wave, simulating perfect collimation. Perfect collimation is not possible in experiments, however. We can simulate the effects of imperfect collimation by instead allowing illumination by a sum of various plane waves, which form a basis for solutions of Maxwell's equations outside the film. The inclusion of this effect into our model is simple, as the transmission curve for multiple incident plane waves is the average of the transmission curves for each individual incident wave (see Appendix I).

We return to the case of square holes in a square lattice. In Fig. 6, we demonstrate the results of averaging transmission curves for various angles of incidence within three different angle intervals about normal incidence: 0.1° , 0.2° , and 1° . The thinner of the two peaks near $\lambda = L$ diminishes in height until all that remains is a single peak, which also decreases in amplitude but at a slower rate. The Wood anomaly remains, causing a minimum in transmission, but this minimum no longer reaches zero transmission.

6. Conclusion

Beginning with a known mode-matching method for evaluating EM scattering from a perforated metal film, we alter the model in several ways. We find that the assumption of perfect conductivity only alters the results minimally in the microwave regime. The assumption of infinite periodicity significantly affects the amount of transmission but does not affect the qualitative shape of the transmission curve, including the Wood anomaly and the location of transmission maxima. Allowing the level of collimation to vary, however, significantly affects the transmission curve, causing it to take on the qualitative features of experiments. For the physical parameters considered in this study, the narrow transmission maximum is reduced

to 50% height when collimation varies by about 0.15 degrees.

Appendix I: The Mode-Matching Procedure

The basic mode-matching procedure may be found in [2, 3]. It is reproduced here in detail with discussion of the alterations used in this report. The beginning of the mode-matching procedure is to write the EM fields to either side of the film as a countable sum of plane waves, and within the film as a countable sum of waveguide modes [2]. Afterwards, we use the matching conditions for Maxwell's equations to solve for the coefficients.

Here, we write out the equations for a two-dimensionally periodic array of holes in a perfect conductor ($|\epsilon_2| = \infty$ inside the metal; $\epsilon = 1$ outside the slab; ϵ_1 inside the holes is finite, typically equal to 1, and μ is identically 1), with the option of multiple holes per period cell (see section 4). We will note along the way the changes required for the case of finite conductivity.

We let the film be periodic in the x - and y - directions, and finite in the z - direction with thickness h . Bloch theory ensures us that the solution of the time-harmonic Maxwell's equations will be Bloch periodic, so that we need only consider a single period cell. Furthermore, full plane wave solutions of Maxwell's equations can be reconstructed if we know only E_x , E_y , and the direction of propagation. Therefore we begin by writing the field left of the slab (in the reflection side) as

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = W_0 e^{ik_{z0}(z+h/2)} + \sum_k r_k W_k e^{-ik_z(z+h/2)}, \quad (6)$$

where W_k represents the x, y components of the plane wave enumerated by k , r_k is the reflection coefficient for that wave, and W_0 is the incident plane wave. The full normalized expression for W_k is

$$W_k = \begin{pmatrix} k_x \\ k_y \end{pmatrix} \frac{e^{i(k_x x + k_y y)}}{\sqrt{L_x L_y |k_t|}}, \quad (7)$$

$$W_k = \begin{pmatrix} -k_y \\ k_x \end{pmatrix} \frac{e^{i(k_x x + k_y y)}}{\sqrt{L_x L_y |k_t|}} \quad (8)$$

for p - and s -polarized waves, respectively, where L_x, L_y are the width of the period cell in the x - and y - directions, $\mathbf{k}_t = (k_x, k_y)^T$, and the values of k_t range over

$$\mathbf{k}_t = \mathbf{k}_{0t} + 2\pi \begin{pmatrix} j_1/L_x \\ j_2/L_y \end{pmatrix} \quad (9)$$

for all integers j_1, j_2 and arbitrary incident wavevector $\mathbf{k}_{0t}$ encoding the angle of incident radiation via $k_{0x} = \omega \sin(\theta_x), k_{0y} = \omega \sin(\theta_y)$. For normal incidence, there is no distinction

between p and s polarization, yet there are two plane waves: $(1, 0)^T \exp(ik_z z)/(L_x L_y)^{1/2}$ and $(0, 1)^T \exp(ik_z z)/(L_x L_y)^{1/2}$.

The x, y components of the electric field of plane waves and waveguide modes are closely related to the x, y components of the magnetic field by

$$\begin{pmatrix} H_y \\ -H_x \end{pmatrix} = \pm Y \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \quad (10)$$

where the constant Y , called the admittance, is equal to ω/k_x for p -polarized plane waves, k_z/ω for s -polarized waves, and whose values for waveguide modes will be detailed below. The sign depends on the direction in which the wave is propagating. Hence, for the regions to either side of the film – the reflection and transmission sides – we may write

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix}_{refl} = W_0 e^{ik_0(z+h/2)} + \sum_k r_k W_k e^{-ik_z(z+h/2)} \quad (11)$$

$$\begin{pmatrix} H_y \\ -H_x \end{pmatrix}_{refl} = Y_{k_0} W_0 e^{ik_0(z+h/2)} - \sum_k r_k Y_k W_k e^{-ik_z(z+h/2)} \quad (12)$$

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix}_{tran} = \sum_k t_k W_k e^{ik_z(z-h/2)} \quad (13)$$

$$\begin{pmatrix} H_y \\ -H_x \end{pmatrix}_{tran} = \sum_k t_k Y_k W_k e^{ik_z(z-h/2)}. \quad (14)$$

The field inside the holes is treated similarly:

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix}_{hole} = \sum_{\alpha, H} M_{\alpha, H} (A_{\alpha, H} e^{ik_{\alpha, H} z} + B_{\alpha, H} e^{-ik_{\alpha, H} z}) \quad (15)$$

$$\begin{pmatrix} H_y \\ -H_x \end{pmatrix}_{hole} = \sum_{\alpha, H} M_{\alpha, H} Y_{\alpha, H} (A_{\alpha, H} e^{ik_{\alpha, H} z} - B_{\alpha, H} e^{-ik_{\alpha, H} z}), \quad (16)$$

where α enumerates the waveguide modes $M_{\alpha, H}$ in hole H , as H ranges over each hole in the basic period supercell. The coefficients $A_{\alpha, H}, B_{\alpha, H}$ are the amplitudes of the mode as it moves in the positive and negative z - directions, respectively.

For rectangular holes in a perfect conductor, the waveguide modes inside the holes for TE and TM polarizations are given by

$$M_{m,n,H} = \left(\begin{array}{c} (-n/b_H) \cos \left(\frac{m\pi}{a_H} (x + a_H/2 - C_{Hx}) \right) \sin \left(\frac{n\pi}{b_H} (y + b_H/2 - C_{Hy}) \right) \\ (m/a_H) \sin \left(\frac{m\pi}{a_H} (x + a_H/2 - C_{Hx}) \right) \cos \left(\frac{n\pi}{b_H} (y + b_H/2 - C_{Hy}) \right) \end{array} \right) / N \quad (17)$$

$$M_{m,n,H} = \left(\begin{array}{c} (m/a_H) \cos \left(\frac{m\pi}{a_H} (x + a_H/2 - C_{Hx}) \right) \sin \left(\frac{n\pi}{b_H} (y + b_H/2 - C_{Hy}) \right) \\ (n/b_H) \sin \left(\frac{m\pi}{a_H} (x + a_H/2 - C_{Hx}) \right) \cos \left(\frac{n\pi}{b_H} (y + b_H/2 - C_{Hy}) \right) \end{array} \right) / N, \quad (18)$$

as m, n range over the nonnegative integers; $(C_{Hx}, C_{Hy})^T$ is the center of hole H ; a_H, b_H are the widths of hole H in the x - and y - directions, and the normalization constant N is given by

$$N = \left(\frac{b_H^2 m^2 + a_H^2 n^2}{4a_H b_H} \right)^{1/2} \quad (19)$$

unless m or n is zero, in which case the 4 in the denominator is replaced by 2. Inside the metal, the waveguide mode is equal to zero. Meanwhile,

$$k_{m,n,H}^2 = \epsilon_1 \omega^2 - \pi^2 \left(\frac{m^2}{a_H^2} + \frac{n^2}{b_H^2} \right), \quad (20)$$

and the admittance $Y_{m,n,H}$ is equal to $k_{m,n,H}/\omega$ for TE modes, and $\epsilon_1 \omega / k_{m,n,H}$ for TM modes.

For circular holes, the modes are expressions involving Bessel functions [12]. For circular holes in an imperfect conductor, the modes are no longer zero inside the metal, and Eq. (10) is replaced with Eq. (5).

It now remains to solve Eqs. (11) - (16) for the transmission and reflection coefficients. The matching conditions for Maxwell's equations tell us that $\mathbf{E}_t := (E_x, E_y)^T$ must be continuous at the interfaces between the slab and the reflection/transmission regions. Thus, for example, at $z = -h/2$, Eq. (11) is equal to Eq. (15). We may solve the resulting equality for r_k by taking the inner product of both sides with W_k^* , integrating over the period cell: the plane waves are orthonormal, and hence will drop out of the equation.

$$r_k = -\delta_{k,k_0} + \sum_{\alpha,H} (W_k^*, M_{\alpha,H}) (A_{\alpha,H} e_{\alpha,H}^{-1} + B_{\alpha,H} e_{\alpha,H}), \quad (21)$$

where $e_{\alpha,H} = \exp(ik_{\alpha,H}h/2)$. Similarly, at $z = h/2$, Eq. (15) is equal to Eq. (13), so that

$$t_k = \sum_{\alpha,H} (W_k^*, M_{\alpha,H}) (A_{\alpha,H} e_{\alpha,H} + B_{\alpha,H} e_{\alpha,H}^{-1}). \quad (22)$$

Meanwhile, for a perfectly conducting metal, $\mathbf{H}_t$ must be continuous at the air-hole interface but may be discontinuous at the air-metal interface. We may enforce this by setting (12) equal to (16) at $z = -h/2$, and (16) equal to (14) at $z = h/2$, and then taking the

inner product with $M_{\alpha,H}^*$ instead of with the plane wave. Since the waveguide mode is zero away from the hole, applying it to both sides of the equation eliminates that region from consideration. We obtain

$$-\sum_k Y_k r_k(M_{\alpha,H}^*, W_k) = -Y_{k_0}(M_{\alpha,H}^*, W_{k_0}) + Y_{\alpha,H} (A_{\alpha,H} e_{\alpha,H}^{-1} - B_{\alpha,H} e_{\alpha,H}) \quad (23)$$

$$\sum_k Y_k t_k(M_{\alpha,H}^*, W_k) = Y_{\alpha,H} (A_{\alpha,H} e_{\alpha,H} - B_{\alpha,H} e_{\alpha,H}^{-1}). \quad (24)$$

For the case of an imperfect conductor, $\mathbf{H}_t$ must also be continuous at the air-metal interface. The same procedure, however, yields this result. We then combine Eqs. (21), (22) with Eqs. (23), (24) to solve for $A_{\alpha,H}$, $B_{\alpha,H}$, and then refer back to Eqs. (21), (22) to find r_k , t_k .

Finally, we calculate transmission by integrating the time-averaged Poynting vector $\frac{1}{2} \text{Re}(E \times H^*)$ over a period cell in the transmission region. Because the plane waves are orthonormal with respect to this integral, it reduces to

$$T = \sum_k \Re(Y_k) |t_k|^2 / Y_{k_0}. \quad (25)$$

This formula implies that the combined transmitted energy of two (or more) incident plane waves may be calculated by finding the transmitted energy of each plane wave separately and then summing the results (as is done in Sec. 5). There is one exception to the previous statement, which is if the two incident waves have wavenumbers which produce the same possible values of k_t via Eq. (9); however, this does not occur in any of the simulations in this study.

Appendix II: Asymptotic Expression for the Propagation Constant

Equation (3) determines the propagation constant k_z for a hybrid waveguide mode in a circular waveguide encased by another material, which each have finite conductivity ϵ_1 and ϵ_2 , respectively. The magnetic constant μ is taken to be the same in both materials.

If the surrounding material has dielectric constant ϵ_2 approaching infinity, then the hybrid mode will approach a TE or TM mode as if the waveguide were surrounded by a perfect conductor. Thus we can solve for k_z asymptotically by allowing k_z to be close to k_0 (the propagation constant for the perfect conductor) and letting $1/\epsilon_2$ be close to zero. The first terms of this asymptotic expansion are

$$k_z \sim k_0 + \frac{C_1}{\epsilon_2^{1/2}} + \frac{C_2}{\epsilon_2}. \quad (26)$$

The constants C_1 and C_2 differ depending on whether the hybrid mode approaches a TE

mode or a TM mode. In the TE case, k_0 is such that $u'_p = a(\omega^2\mu\epsilon_1 - k_0^2)^{1/2}$ is the p^{th} zero of the Bessel function derivative J'_n , and

$$C_1^{TE} = \frac{-J_n(u'_p)\alpha}{a(-\omega^2\mu)^{1/2}J''_n(u'_p)k_0} \left(1 + \frac{n^2k_0^2}{a^2\alpha^2}\right) \quad (27)$$

$$C_2^{TE} = \frac{-J_n(u'_p)\alpha}{k_0J''_n(u'_p)} \left(1 + \frac{n^2k_0^2}{a^2\alpha^2}\right) \left[\frac{1}{a^2\omega^2\mu} \left(\frac{1}{2} + \frac{n^2k_0^2}{a^2\alpha^2} + \frac{J_n(u'_p)n^2}{J''_n(u'_p)a^2\alpha} \left(2 + \frac{4k_0^2}{\alpha}\right) \right) \right], \quad (28)$$

where $\alpha = \omega^2\mu\epsilon_1 - k_0^2 = (u'_p/a)^2$. Meanwhile, in the TM case, k_0 is such that $u_p = a(\omega^2\mu\epsilon_1 - k_0^2)^{1/2}$ is the p^{th} zero of J_n , and

$$C_1^{TM} = -\frac{\epsilon_1(-\omega^2\mu)^{1/2}}{ak_0} \quad (29)$$

$$C_2^{TM} = \frac{\epsilon_1\omega^2\mu}{2k_0} \left(\frac{1}{a^2k_0^2} - \frac{1}{a^2\beta} + \frac{J''_n(u_p)}{J'_n(u_p)a(\beta^{1/2})} \right) + \frac{\epsilon_1}{2a^2k_0}, \quad (30)$$

where $\beta = \omega^2\mu\epsilon_1 - k_0^2 = (u_p/a)^2$.

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List of Figure Captions

Fig. 1. A plane wave is incident on a metal film of thickness h , with holes of width (in the case of rectangular holes) or diameter (in the case of circular holes) w periodically placed with period L . Some of the incident energy is reflected, and some enters the hole, exiting through to the other side.

Fig. 2. (a) Transmission profile for a normally incident plane wave on a 2-dimensionally periodic array of square holes. The period $L = L_x = L_y$ is arbitrary; the side lengths of the holes is $a = 0.4L$ and the thickness of the metal is $h = 0.2L$. The number λ is the wavelength $2\pi/\omega$. There are two Wood anomalies present at $\lambda/L = 1$ and $\lambda/L = (\frac{1}{2})^{1/2}$. (Inset) Closeup of the double maximum. (b) The incoming plane wave is incident at a 5 degree angle. New Wood anomalies have branched off from the ones already present. The double maximum remains but shifts to the right. (Inset) Closeup of the double maximum.

Fig. 3. Same parameters as Fig. 2 for the case of finite conductivity. The solid line is for $\epsilon_2 = -7 \cdot 10^4 + i5 \cdot 10^7$, a typical value of the dielectric constant of copper in the microwave range [11]. The values for Al, Cu, and Ag do not differ enough to produce visible change in the transmission curve. (In reality, the value of ϵ_2 changes with λ , but again this dependence does not produce any visible difference.) The transmission profile is nearly indistinguishable from the perfect conductor case. The dotted line is for $\epsilon_2 = -1.7 \times 10^4 + i2 \times 10^4$, a typical value of the dielectric constant of silver in the infrared range [11]. The primary difference is that the height of the narrow peak has decreased to near 90%. (Inset) Closeup of the narrow maximum.

Fig. 4. Same as Fig. 3 produced by COMSOL Multiphysics. (a) A plane wave is incident from one boundary and transmitted energy is calculated at the other. Because we assume normal incidence, periodic boundary conditions are enforced on the four sides of the period cell. (b) Two different films are simulated, one that is a perfect electric conductor (PEC), and another which is Al at a frequency of 10GHz. The transmission profiles of both metals agree. The Wood anomaly and the narrower transmission maxima, however, are not captured exactly.

Fig. 5. Normalized-to-area transmission profile for a plane wave normally incident on a one-dimensionally periodic array of slits in a perfectly conducting film, with arbitrary distance L between neighboring holes, width $0.4L$ and film width $0.2L$. (a) Basic case with an infinitely periodic array of slits. (b) We have placed spacing of $90L$ between every block of 10 slits. (c) We have placed spacing of $490L$ between every block of 10 slits. The transmission profile decreases somewhat in magnitude, and Fabry-Perot-like oscillations appear due to

the interference of separate 10-hole blocks. Aside from this, the shape of the curve does not change.

Fig. 6. Transmission profiles for square holes periodically repeated with period L and hole width $a = 0.4L$ in a perfectly conducting film of width $h = 0.2L$, illuminated by plane waves at various angles. (a) Incident waves vary from -0.1° to $+0.1^\circ$ in both the x - and y -directions. (b) Incident waves vary from -0.2° to $+0.2^\circ$. (c) Incident waves vary from -1° to $+1^\circ$. The thinner of the two transmission maxima diminishes in height until it is no longer apparent, and the single remaining maximum decreases in magnitude more slowly.

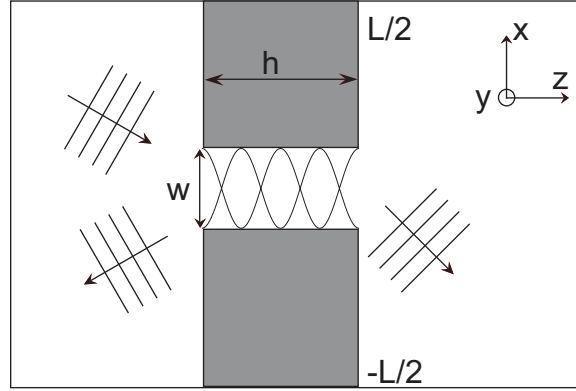


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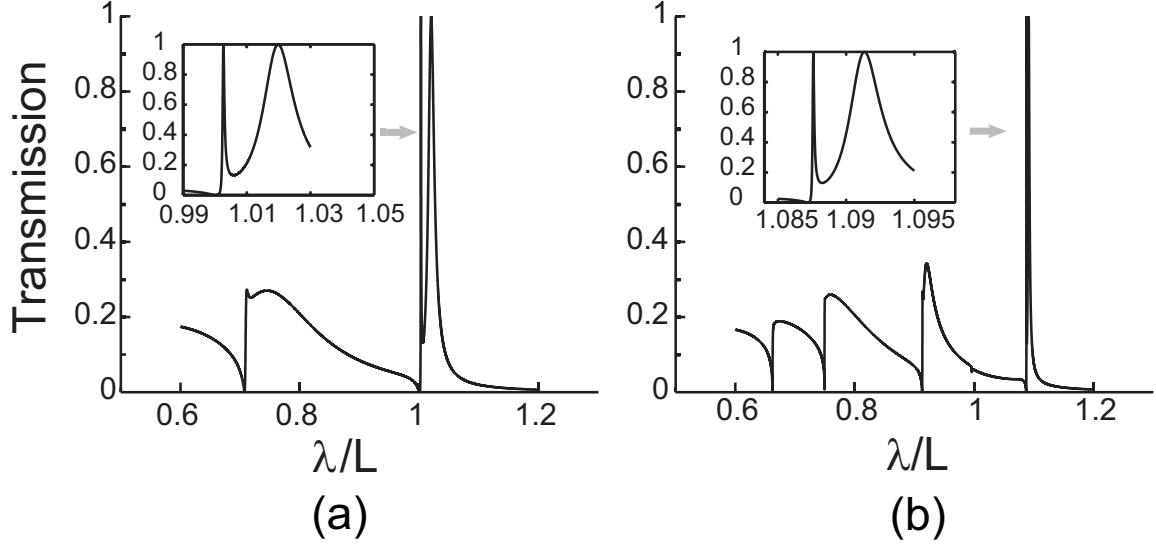


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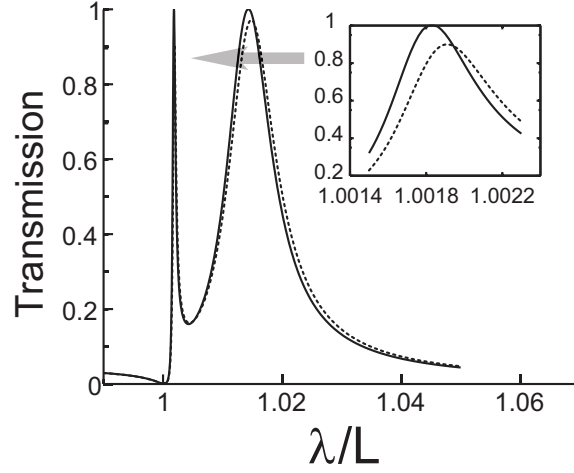


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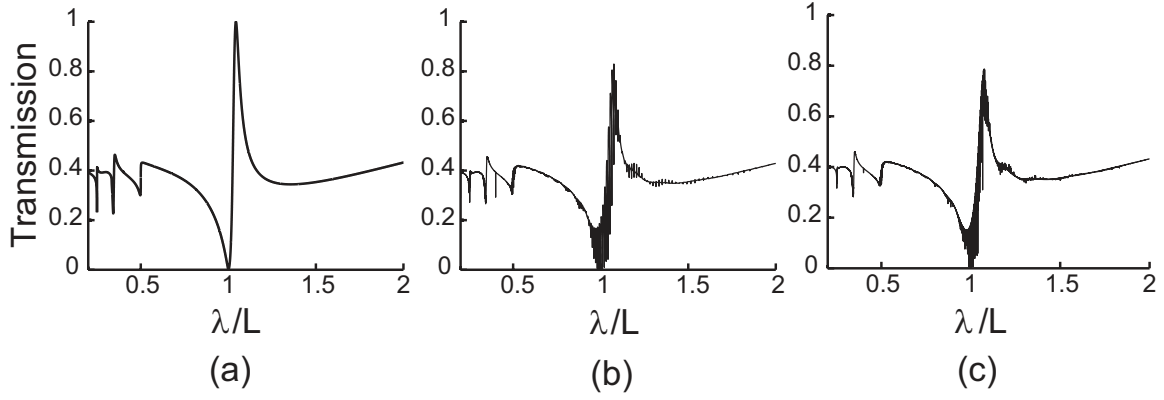


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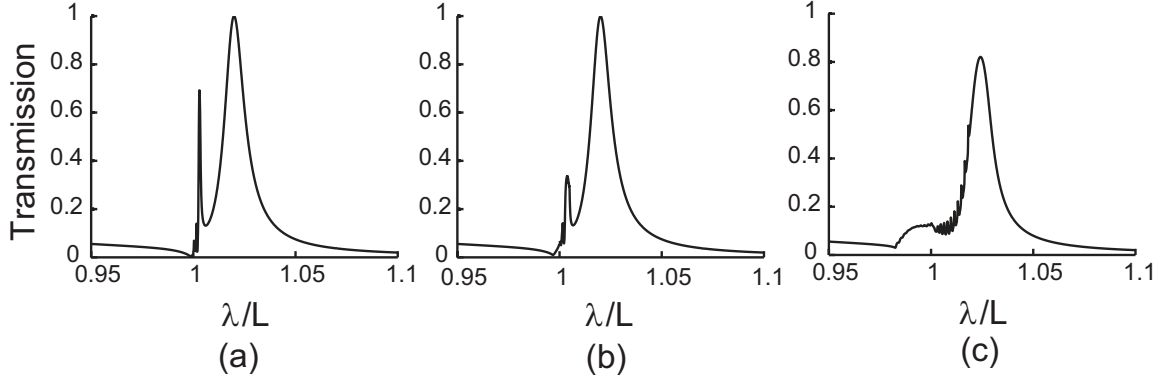


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